NAVGURUKUL HACKATHON — CHALLENGE 3

Learning Recommendation System – Project Report

1. Introduction

The goal of this project is to build an end-to-end learning recommendation system that predicts the next learning item (skill/exercise) a student is likely to interact with, based on their historical activity.
This aligns with Challenge 3 requirements: data preprocessing, temporal splitting, baseline models, advanced sequence models, evaluation metrics, and a demo application.

We used the **ASSISTments Skill Builder dataset**, which contains ~700k student interactions with 100 unique learning skills.

Dataset Link: https://sites.google.com/site/assistmentsdata/datasets/2015-assistments-skill-builder-data?utm_source=chatgpt.com

2. Dataset and Preprocessing

Dataset Structure

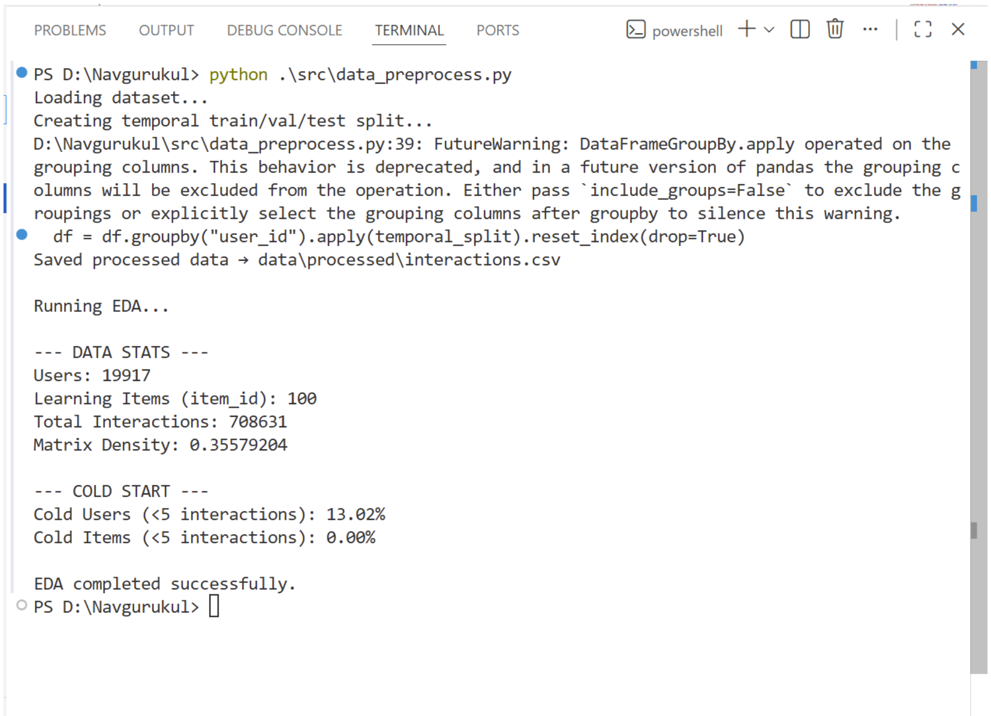
Each row represents one student–problem interaction:

- user_id — unique student
- item_id — learning skill/problem
- correct — success indicator (0/1)
- ts — timestamp / interaction order
- split — train/val/test assignment

Data Cleaning

- No missing values in key fields
- No duplicated interactions
- Correctness values only {0,1}
- Sorted each student's history chronologically

EDA Highlights



```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                    powershell
PS D:\Navgurukul> python .\src\data_preprocess.py
Loading dataset...
Creating temporal train/val/test split...
D:\Navgurukul\src\data_preprocess.py:39: FutureWarning: DataFrameGroupBy.apply operated on the
grouping columns. This behavior is deprecated, and in a future version of pandas the grouping c
olumns will be excluded from the operation. Either pass `include_groups=False` to exclude the g
roupings or explicitly select the grouping columns after groupby to silence this warning.
  df = df.groupby("user_id").apply(temporal_split).reset_index(drop=True)
Saved processed data → data\processed\interactions.csv

Running EDA...

--- DATA STATS ---
Users: 19917
Learning Items (item_id): 100
Total Interactions: 708631
Matrix Density: 0.35579204

--- COLD START ---
Cold Users (<5 interactions): 13.02%
Cold Items (<5 interactions): 0.00%

EDA completed successfully.
PS D:\Navgurukul>
```

3. Baseline Models

3.1 Most Popular Model

Ranks items by global frequency and recommends the top items a student has not attempted yet.

- Simple, fast baseline
- Cold-start friendly

3.2 Item-Based Collaborative Filtering

Computes cosine similarity between items based on shared users.

Steps:

1. Build item→user interaction sets
2. Compute item-item similarity
3. Recommend items similar to the student's history

CF personalizes recommendations but does not model time or sequence.

4. Advanced Model — GRU Sequence Recommender

The advanced model is a GRU-based sequential recommender, which learns transition patterns in students' learning sequences.

Model Architecture

- Embedding layer for item representations
- GRU (Gated Recurrent Unit) for sequence modeling
- Fully connected layer predicting next item
- CrossEntropyLoss for next-item prediction

Why GRU?

- Captures order of learning events
- Learns long-term dependencies
- Much stronger than static baselines
- Satisfies Challenge 3 requirement for sequence models (RNN/Transformer/SASRec)

5. Evaluation Metrics

We evaluated all models using standard ranking metrics:

- **Precision@10**
- **Recall@10**
- **NDCG@10**
- **MRR@10**

These metrics measure accuracy, ranking quality, and model usefulness.

Model	Precision@10	Recall@10	NDCG@10	MRR@10
Most Popular	0.015	0.022	0.021	0.044
Collaborative Filtering	0.033	0.051	0.055	0.120
GRU Sequence Model	0.138	0.344	0.440	0.872

Interpretation

- CF improves over Most Popular by leveraging co-occurrence patterns.
- GRU outperforms both baselines by a large margin because it models temporal learning behavior, not just item similarity.
- GRU's **MRR=0.87** indicates it ranks the correct next skill very high consistently.

UI: